

ERP WEEK 2

Week 2 — Session 3

Teaching Script

Session 3: ERP Architecture
Session 4: Core ERP Modules

ERP Architecture: Client–Server, Cloud, and Hybrid Models

Session Opening

In our previous ERP lessons, we established that enterprise resource planning connects departments, transactions, and shared information and we also tackle about the evolution of it that how ERP evolve in time so nag evelove sya in 5 stages which is ang MRP, MRP II, ERP, ERP II, and CLOUD ERP.

Today, we'll proceed na to the technical side. If Finance, Procurement, Inventory, Human Resources, and other functions are connected, unsa man ang structure behind that connection?

- Unsaon pag-travel sa user request sa sulod sa system?
- Asa gi-apply ang business rules ug asa gi-store ang data?
- Ug asa man dapat i-run ang ERP—on-premise, sa cloud, o hybrid?

Importante ni nga mga pangutana kay ang ERP dili lang kay software nga i-install ug kalimtan. Ang architecture mo-affect sa speed, security, scalability, ug sa responsibilidad sa kompanya. Kung bati ang architecture decision, maglisod tag manage sa system. Pero kung sakto, structured ug reliable atong operations.

Mao nga for this session, mao ni atong buhaton:

1. Tun-an nato ang three-tier ERP structure: presentation, application, ug data layer.
2. Atong i-trace ang client-server request gikan sa UI padung sa database ug pabalik.
3. I-compare nato ang on-premise, cloud, ug hybrid deployments.

Huna-hunaa ninyo na architecture is technical, but its purpose is organizational. Gi-design ni para sa controlled processes ug trustworthy data.

Sige, let us begin with Session 3.

Slide 1 - ERP Architecture: Client–Server, Cloud, and Hybrid Models

Stage Directions

[Shows Slide 1]

[Waits for the title animation]

[Instructor reads slide]

Verbatim Teacher Script

Week 2, Session 3: ERP Architecture - Client-Server, Cloud, and Hybrid Models.

How ERP layers, infrastructure, and deployment choices support integrated organizational operations.

Kini nga title naghatag nato og two levels of discussion.

First is architecture—ang internal structure sa ERP system. Kini nag-tubag sa mga pangutana sama sa: Asa dapit makita sa user? Asa gi-apply ang business rules? Ug asa gi-store ang records?

Second is deployment—kung asa i-run ang components ug kinsa ang mag-manage niini. Gi-host ba sa kompanya? Cloud service ba? O hybrid system?

Usahay, mag-assume ang mga estudyante nga lahi kaayog basic logic ang cloud ERP kumpara sa on-premise. Dili necessarily. Pareho ra gihapon silang naggamit og presentation, application, ug data layers. Ang dako nga kausaban kay giunsa pag-distribute ang infrastructure ug management responsibilities.

Throughout this session. Atong i-connect ang layers, the request flow, and the deployment choices. By the end of this class, makasabot na mo dili lang sa components, kundi why the organization selected that structure ug unsa ilang responsibilidad.

Transition

Before nato i-examine ang layers, let us make our learning goals clear para kabalo mo unsay paminawon sa discussion.

[Clicks next slide]

Slide 2 - Intended Learning Outcomes

Stage Directions

[Shows Slide 2]

[Reveals each learning outcome one at a time]

[Instructor reads slide]

Verbatim Teacher Script

By the end of this session, you should be able to:

describe the presentation, application, and database tiers;

explain how requests move through a three-tier system;

differentiate on-premise, cloud, and hybrid deployment models;

and **analyze** architecture choices using security, performance, scalability, integration, and governance considerations.

Aron mas simple, mao ni atong pasabot sa practical terms:

First, dapat mailhan ninyo ang responsibility sa kada tier. Ang presentation tier kay asa mo-interact ang user. Ang application tier ang nag-handle sa business logic. Ug ang data tier ang nag-maintain sa shared records.

Ikaduha, you should be able to trace a transaction. Kung mag-submit og request ang user, asa man na moagi? Which layer validates it? Unsaon pag-involved sa database, ug unsaon pagbalik sa data ngadto sa user? We will trace that movement step-by-step.

Ikatulo, i-compare nato ang deployment models using practical trade-offs. Dili lang kay 'cloud is online' or 'on-premise is local.' Dapat masabtan ninyo kinsa ang mag-manage sa infrastructure, unsa ka paspas mo-scale, ug asa nakatutok ang governance.

Sa katapusan, we will analyze architecture considerations. Ang security, performance, scalability, integration, ug governance kay connected. Pananglitan, kung magdungag tag external integrations, mo-improve atong processes pero modako pod atong security and ownership requirements.

Mao nga while I discuss, paminawa ang connection sa matag concept para makita ninyo how the whole architecture supports integrated operations.

Transition

Aron masabtan nato ngano importante ni, atong tan-awon unsa gyuy mapuslan sa architecture sa operations sa ERP.

[Clicks next slide]

Slide 3 - Why Architecture Matters in ERP

Stage Directions

[Shows Slide 3]
[Reveals the four cards in sequence]
[Points to each card as it appears]
[Instructor reads slide]

Verbatim Teacher Script

Architecture turns ERP integration into an operational system. Dili lang ni technical diagram; it connects people, processes, data, controls, and goals. Dynamic kini sa upat ka areas:

1. **User Access:** Dili pareho ang makita sa procurement officer, warehouse staff, ug manager. The architecture supports **role-based access** para controlled ang pag-interact sa system.
2. **Process Execution:** Dili lang kay tig-store og final results ang ERP. Kini ang nag-coordinate sa validations, approvals, ug calculations. *Pasabot ani*, the system knows what rules to apply and what to do if there is an error.
3. **Data Integrity:** Ang procurement ug accounting naggamit og parehong supplier or item record. A shared database keeps data consistent so relationships among records remain reliable.
4. **Operational Continuity:** ERP supports daily operations. Kung mag-down ang system, delayed ang inventory, mohunong ang purchasing, ug mawad-an og visibility ang mga managers.
Mao ni ang importanteng connection: architecture creates conditions for integrated work.

Imagine a multi-branch retail company. Kung bati ang architecture, usa ka branch hinay kaayog screens, ang pikas delayed ang transactions, ug ang management makakitag lahi-lahi nga stock figures. It affects purchasing, customer service, and financial reporting.

So when evaluating architecture, always ask: *Makatabang ba ni sa kompanya para mabuhat ang trabaho reliably and under control?*

Transition

Karon nga kabalo na ta nganong architecture matters, let us examine the basic three-tier structure and see how interaction, processing, and storage are separated.

[Clicks next slide]

Slide 4 - ERP Architecture as a Layered System

Stage Directions

[Shows Slide 4]

[Opens the enlarged diagram]

[Moves the cursor from the top layer downward, then follows the arrows upward]

[Instructor reads slide]

Verbatim Teacher Script

ERP architecture separates interaction, processing, and storage into three layers. At the top is the presentation tier for user screens. In the middle is the application tier for business rules and workflows. At the bottom is the data tier for the shared database. Each tier has a distinct responsibility.

Atong i-break down ang diagram from top to bottom.

The presentation tier is ang part nga makita ug ma-experience sa user. Mapasulod man og quantity o mag-click og submit, the interaction begins here.

Ang request then moves downward to the application tier. This is where the system interprets the request in business terms. Pananglitan: Is the user authorized? Is approval required?

Ang application tier ang nag-coordinate ani nga rules. If it needs stored information, kini makig-communicate sa data tier to retrieve or update records, like supplier data or item balances.

Karon, pag abot sa **data tier** atong sundon ang arrows upward. the database dre ma stored ang mga data or informations sa mga transaction then eh returns the result to the application tier, which processes the response. Then, ang presentation tier na pod ang mo-display sa result ngadto sa user—pananglitan, a confirmation number or an error message.

Para mas masabtan, let's use a simple university example:

Presentation Tier: Usa ka staff member mag-select og lab supplies ug mag-input og quantity.

Application Tier: I-validate sa system kung authorized ba ang user ug checks the request against university policy.

Data Tier: The database pulls the records and stores the approved transaction.

Upward Flow: The system updates the screen, ug makita sa staff nga submitted na ang request.

Nganong importante man nga separated kini nga responsibilities?

First, it's easier to manage: Pwede nato usbon ang interface nga dili maapektuhan ang database.

Second, it supports security: Kay dili man direkta nga maka-manipulate ang users sa database.

Third, it helps with scalability and consistency: Rules are applied centrally, making the system more stable.

Huna-hunaa ninyo, these three tiers are logical responsibilities. Pwede sila mag-run sa different physical servers, pero dili pasabot nga kinahanglan gyud og tulo ka physical machines. The key idea is the separation of function, dili ang gidaghanon sa boxes sa server room.

Mao nga ang downward direction represents the request, ug ang upward direction is the result. Nakuha ra?

Transition

Keep the request-and-response flow in mind. We will start with the presentation tier, then move inward to the application and data responsibilities.

[Clicks next slide]

Slide 5 - Presentation Tier: Human–System Interaction

Stage Directions

[Shows Slide 5]

[Reveals the definition, then the four cards]

[Points to the Access channels and Role-based views cards]

[Instructor reads slide]

Verbatim Teacher Script

The **presentation tier** handles human-system interaction. It is the visible interface where authorized users enter, review, and act on ERP information. It supports access channels, basic input validation, role-based views, and usability.

Mao ni ang layer nga tawgon sa mga employees og "*the ERP system*" kay mao man ang ilang makita sa screen. Pero remember: the screen is just the front door. It presents info and receives input, pero dili kini ang nag-process sa tibuok workflow.

Let's break down its key elements:

- **Access Channels:** Kini ang controlled entry points. Users can access the ERP through a web browser, mobile app, desktop client, or warehouse kiosk. Depende kini sa imong role ug working environment.
- **Input Validation:** automated checks direct to the interface to prevent errors. Pananglitan, dili ka kapadayon kung blank ang required fields, or mapugngan ang pag-type og letters sa numeric fields. Basic ra kini nga check, ha? Ang screen can check if a quantity was entered, pero the application tier dictates if allowed ba nga quantity.
- **Role-Based Views:** dynamic ang menus ug fields depende sa imong assigned responsibility. Dili tanan user makakita sa tanan data. A requester can create a purchase request, pero dili siya ang maka-approve sa iyang kaugalingong request.
- **Usability:** Ang nindot na interface prevents mistakes. Clear navigation and helpful messages help users perform the correct action. Dili lang sya nga design lang, design sya for easy to understand sa mga users sa pag gamit sa system; kay it helps with process in terms of using the system

Balikan nato atong example.

The staff sees a form with fields for department, item, and quantity. Pwede naay features automated based sa user profile.

When they click Submit, the presentation tier packages that input ug ipadala dayon kini ngadto sa application tier.

Unsa man ang dili dapat mahitabo? The user interface must never directly manipulate the database without validation. Dili sad pwede nga naay other business rules dynamic sa web, desktop, or mobile versions.

The presentation tier has a clear but limited responsibility: to make authorized ERP work visible, understandable, and actionable. Ang next layer na ang mag-decide kung unsay sunod buhaton sa request.

Transition

Na-identify na nato kung unsa ang makita ug i-enter sa users. Now let us examine the layer that coordinates rules, workflows, calculations, security, and transaction control.

[Clicks next slide]

Slide 6 - Application Tier: Business Logic and Coordination

Stage Directions

[Shows Slide 6]

[Reveals the definition]

[Reveals the six cards in two groups]

[Moves the cursor across the cards to show how they work together]

[Instructor reads slide]

Verbatim Teacher Script

The **application tier** handles business logic and coordination. It interprets user requests, applies rules, manages workflows, performs calculations, and coordinates integrations.

Atong sundon kung unsa ang buhaton sa application tier kung naay moabot nga request:

1. **Interprets the request:** If ever nay transactio ang user and mag input na sya sa mga ipang input niya then after user clicks submit, ug sabton dayon sa system ang transaction event. Kung this is a purchase request, an inventory transfer, or an invoice?
2. **Applies Business Rules:** I-verify sa system ang mga policy. Sa atong example, active ba ang department? Allowed ba ni nga item? Reasonable ba ang quantity? So dre sya eh validate
3. **Workflow Services:** I-determine kung kinsa ang sunod nga mo-approve. Mapunta ba ito sa department head, budget officer, or procurement staff?
4. **Integration Services:** Pinaagi sa APIs (Application Programming Interface), kini makig-communicate sa laing modules, devices, or external banking and supplier portals.
5. **Calculations:** Gina-compute ang pricing, taxes, ug costing. Centralized kini para consistent ang computation, gamit man nimo ang mobile o desktop screen.
6. **Security Enforcement:** I-verify niini kung authorized ba gyud ang user. Kung dili hiding a button on the screen is not enough; the application tier must block unauthorized access directly.
7. **Transaction Control:** Kung ma-approve na ang purchase request, i-execute sa system ang connected nga mga updates sa database. Pananglitan, (1) kuhaan ang budget sa department ug (2) dugangan ang pending orders sa inventory. Pinaagi sa transaction control, dapat dungan kini nga ma-save. Kung naay error sa system samtang nag-update sa inventory, i-reverse/i-atras usab ang pagkuha sa budget aron dili magka-bungi o maguba ang accounting records sa system.

Diri makita ang connection sa daily operations. Control is maintained through **transaction control**. Kinahanglan magdungan og ka-update ang related records—sama sa pagsaka sa inventory quantity ug ang iyang financial effect. Kung ma mali ang isa, i-roll back sa system ang tibuok transaction para dili magkagubot ang data.

Mao ni ang key idea: the application tier is the coordinator. logical responsibilities ni, ha? Kay it handles the workflow, rules, and security.

So basig mag ask mog question: "*Mag-store ba og data ang application tier?*" So the answer is no, Kay ang Application tier is it's hold data temporarily during processing, pero ang long-term, durable enterprise records belong to the data tier.

Transition

Nasabtan na nato ang role sa application tier as the coordinator. The next question is what kinds of records the ERP must preserve and how the data tier protects them.

[Clicks next slide]

Slide 7 - Data Tier: Shared Enterprise Memory

Stage Directions

[Shows Slide 7]

[Reveals the left panel bullets]

[Moves the cursor to the four data categories on the right]

[Instructor reads slide]

Verbatim Teacher Script

The **data tier** serves as the *shared enterprise memory*. Its job is to store one controlled source for enterprise records, preserve relationships, and protect data integrity.

Dili lang kini basta-basta folder sa mga files, ha? It organizes data into four main ERP categories:

- **Master Data:** These are stable reference records reused across processes. Pananglitan, suppliers, customers, items, accounts, and employees. If a supplier appears in Procurement and later in Accounts Payable, both modules refer to this same master record.
- **Transactional Data:** Kini nag-record sa business events over time. Pre-loaded with dates, quantities, and statuses. Examples include requisitions, purchase orders, invoices, or payments.
- **Configuration Data:** This defines how the ERP is organized and how it behaves. Diri gi-set ang company structures, posting rules, numbering sequences, and approval thresholds. Isang ka gamay lang nga configuration kay it changes a big operational effect.
- **Audit Data:** This records who changed what, when, and how. It supports traceability. Kung naay giusab sa supplier bank details, kinsa ug kanus-a man kini gi-update? Dynamic kini for transparency and internal control.

Atong i-connect ang four categories sa atong example:

The item and supplier are **master data**. The purchase request is **transactional data**. The department budget limit and approval threshold are **configuration data**. Ang log kung kinsa ang nag-approve ug kanus-a is **audit data**.

Ang **data tier** nagaprotekta usab sa **integrity** pinaagi sa mga **technical controls** sama sa mga **keys and constraints**. Pananglitan, ang usa ka **transaction should never refer to a non-existent supplier** (dili gyud pwede nga mag-refer ang transaction sa supplier nga wala nag-exist sa system). Gawas niana, ang **backup and recovery** nagaprotekta sa mga **records from system failures** (aron dili mawala ang data kung mag-crash ang system). Hinumdumi nga ang **centralized memory is only valuable when it is completely trustworthy** (walay pulos ang centralized memory kung dili kini hingpit nga kasaligan)

Transition

Let us now trace one complete request from the client interface to application processing, then to the database, and finally back to the user.

[Clicks next slide]

Slide 8 - Client–Server ERP Model

Stage Directions

[Shows Slide 8]

[Opens the enlarged diagram]

[Points to Client layer, Application server, and Database server]

[Moves the cursor along the request arrows and then along the response arrows]

[Instructor reads slide]

Verbatim Teacher Script

This model shows how requests move from users to services, down to the database, and back. The **client layer** handles the user interface, the **application server** processes the business logic, and the **database server** maintains the shared records.

Atong sundon ang dynamic communication flow step-by-step using our university example:

1. **User Action:** The authorized staff member enters the lab supply request in the client layer and clicks Submit.
 2. **Client Request:** Ang client application collects the data and sends a request to the application server. Remember, the client layer cannot directly change enterprise records.
 3. **Application Processing:** The application server interprets the request. I-check niini ang user authorization, applies business rules, and determines the workflow path.
 4. **Database Query/Update:** Kinahanglan og data sa application server, so it queries the database server. If valid ang transaction, the database writes the controlled record and status.
 5. **Returned Data Result:** Ang database server returns the raw result or confirmation back to the application server. The application server interprets the result and prepares a business response.
 6. **Displayed Response:** The application sends the response to the client layer, ug makita na sa screen sa user ang confirmation number or pending approval status.
- Notice ninyo the flow: each layer performs its own function.

Kung naay mangutana nganong "*Hinay ang ERP*," technical teams must trace this exact flow. The delay could be from the client network, overloaded application processing, or an inefficient database query. The arrows also represent trust boundaries—every step must be authenticated and logged.

Mao ni ang complete pattern: action, request, processing, query, result, and display. Nakuha ra ang sequence?

Transition

The structure is clear. Let us now compare deployment environments, beginning with an ERP system operated within organization-controlled infrastructure.

[Clicks next slide]

Slide 9 - On-Premise ERP Deployment

Stage Directions

*[Shows Slide 9]
[Reveals the definition]
[Reveals the four cards]
[Highlights Organizational control and Internal responsibility]
[Instructor reads slide]*

Verbatim Teacher Script

On-premise ERP means organization-managed infrastructure. The system is installed and operated within servers and networks controlled directly by the company or its local hosting provider.

Diri sa on-premise we have 2 ka important questions ang atong tubagon:

- **Where does the system run?** So naga run ang system sa inyong kaugalingong server room o data center. The hardware, storage, and network are under your direct technical staff.
- **Who manages it?** Ang inyong internal IT team. Sila ang bahala sa capacity planning, monitoring, patching, backups, and physical security.

The biggest strength of on-premise deployment is **direct control**. Ang organization has the full freedom to schedule updates, connect specialized local devices, and configure infrastructure according to unique operational needs.

Sa practical side, mao ni ang pait nga kamatuoran: direct control also means **direct responsibility**. Kinahanglan ninyo mag-invest sa maintenance, staff training, ug regular upgrades. Dili pwede nga gusto ta og full control pero walay klarong ownership sa patching, system backup, and disaster recovery.

Naay misconception nga mas secure ang on-premise. It can be secure, pero only if the IT team has the discipline and skills to maintain it. A neglected local server can actually be more vulnerable than a professionally managed cloud service.

So, remember the trade-off: On-premise provides maximum local control, but it carries a heavy operational and financial management burden.

Transition

On-premise places platform operation largely under the organization. Let us now examine cloud ERP, where more of the technical platform is delivered and maintained by a provider.

[Clicks next slide]

Slide 10 - Cloud ERP Deployment

Stage Directions

[Shows Slide 10]

[Opens the enlarged responsibility diagram]

[Points first to Organization, then Cloud ERP provider, then the shared-responsibility statement]

[Instructor reads slide]

Verbatim Teacher Script

Cloud ERP means provider-managed service delivery. The system runs on infrastructure hosted by the provider and is accessed through a secure network connection.

Diri sa cloud, implementation follows the principle of **shared responsibility**:

- **The Provider manages the platform:** Sila ang bahala sa hardware, database platform, updates, security operations, backups, and system availability.
 - **The Organization manages the use:** Kamo gihapon ang responsable sa user access, role assignments, process configurations, and data governance.
- The main advantage of cloud deployment is **operational focus and speed**. Mas paspas kini i-provision, scalability is easier without buying new hardware, and updates are delivered regularly.

Pero huna-hunaa ninyo, cloud does not mean zero IT work or that the provider runs your business. Pananglitan, for a regional hotel group: the provider maintains the platform, pero ang hotel group gihapon ang mag-decide kinsang employee ang pwede mo-approve og purchases o unsaon pag-structure ang data. If you give the wrong person payment authority, or if magkapareha ang inyong data due to duplicate entries, dili kana sala sa cloud provider. Data discipline relies entirely on you.

Ang dako nga dependency sa cloud is **connectivity**. You need a reliable internet connection. Plus, dynamic changes happen to your IT team's workload. Instead of fixing hardware, your team will focus on vendor oversight, data protection, and configuration management.

Maong remember lang guys dre sa cloud: The provider controls the platform, but the organization controls the use, roles, processes, and data quality.

Transition

Cloud moves many platform responsibilities to a provider. Some organizations, however, cannot or do not want to move every component at once. That leads us to hybrid ERP.

[Clicks next slide]

Slide 11 - Hybrid ERP Deployment

Stage Directions

[Shows Slide 11]

[Opens the enlarged hybrid diagram]

[Points to the on-premise environment, then the integration layer, then the cloud environment]

[Moves the cursor along the two-way arrows]

[Instructor reads slide]

Verbatim Teacher Script

Hybrid ERP is the combination of coordinated on-premise and cloud components. Ang on-premise side, usually, siya ang naga-handle sa core records or specialized local device integration, habang ang cloud side naman runs hosted services like analytics.

The key word here is **coordinated**. Ang hybrid, dili lang kay basta-basta '*naay systems sa local, naay systems sa cloud.*' Dynamic data must exchange securely through an integration layer using APIs and middleware. Kung naay mahitabo nga transaction, ang integration layer ang mag-coordinate sa flow:

1. **Data Transfer:** Ang local production system mag-record sa packed tuna boxes, tapos ipadala niana pinaagi sa middleware ngadto sa cloud ERP inventory module.
 2. **Sync and Timing:** Dili tanan integrated system kay real-time, ha? Ang uban naga-move via scheduled batches. The company must define acceptable timing para magpabilin nga accurate ang inventory levels.
 3. **Error Handling:** Kung mag-drop ang network ug mapadpad o mapakyas ang message, ang integration layer dapat makasalo sa error. It must retry and notify the team para dili magka-conflict ang data.
 4. **Identity Management:** Kung naay worker nga muresign o mugawas sa kompanya, central coordination must deactivate their access sa local systems ug sa cloud systems sa parehong higayon.
- Hybrid deployment is very practical because it gives you flexibility and lets you migrate gradually. Pero ang trade-off ani kay complexity. More interfaces mean more failure points and tighter governance requirements.

In a hybrid model, do not just ask, '*Can these systems connect?*' Pangutan-a pod: Which system is the ultimate source of truth? How quickly should data synchronize? And what happens when a message fails?"

Transition

"Karon, atong i-compare ang on-premise, cloud, and hybrid gamit ang parehong decision dimensions: ownership, speed, control, scalability, connectivity, and management focus.

[Clicks next slide]

Slide 12 - On-Premise, Cloud, and Hybrid Comparison

Stage Directions

[Shows Slide 12]

[Checks whether the table is readable]

[Highlights one row at a time instead of reading every cell]

[Instructor reads slide]

Verbatim Teacher Script

Deployment models distribute control and responsibility differently. No model is universally superior, ha? Ang iyahang fit mag-depende gyud sa organizational goals, resources, ug governance capacity sa kompanya.

Atong i-compare ang models by looking at these key decision dimensions:

- **Infrastructure Ownership:** Sa On-premise, organization-managed gyud na. Kung sa Cloud, provider-managed na siya, while ang Hybrid shares the responsibility. Kani nga dimension ang mag-shape sa inyong tibuk IT budget ug staffing requirements.
 - **Deployment Speed:** Siyempre, mas paspas ang Cloud kay existing na man ang platform. Ang On-premise, medyo dugay-dugay kay mag-setup pa mo sa local hardware. Ang Hybrid speed naman mag-depende sa integration complexity—sayon tan-awon sa sugod, pero dugay kaayo i-test.
 - **Customization Control:** On-premise offers full freedom, pero kamo pod ang dapat mag-maintain sa inyong gipang-build. Ang Cloud, restricted mo sa limits ug update policies sa provider. Ang Hybrid offers selective control, apan ang mga custom links mag-add og dako nga overhead.
 - **Scalability:** Sa On-premise, kailangan og manual hardware planning ug bug-at nga investment. Ang Cloud kay elastic, dali ra kaayo mag-scale within your subscription limits. Ang Hybrid naga-gamit og mixed scaling, meaning bantayan gyud ninyo ang mga system bottlenecks.
 - **Connectivity Dependency:** On-premise relies heavily on internal local networks. Ang Cloud naman, dako kaayo og dependence sa stable nga internet connection. Ang Hybrid, dependent sa duha—kung mag-drop gani ang connection in between, paktay, maputol pod ang synchronization.
 - **Management Focus:** On-premise focuses on physical infrastructure maintenance—atimanon gyud ninyo ang hardware. Sa Cloud, ma-shift inyong attention ngadto sa service governance, data quality, ug vendor management. Sa Hybrid, grabeng tight integration ownership ang ginademand ani.
- Let's apply this to actual scenarios para mas masabtan ninyo:
- Kung ang usa ka government office naay stable nga local data center ug strict kaayo ang security restrictions, On-premise or Hybrid may be the best fit for them.
 - Kung growing multi-branch SME naman nga gusto og paspas nga deployment pero minimal lang ang IT staff, dako kaayo og bentahe ang Cloud.
 - Pero kung manufacturer mo nga naay mga local factory equipment pero gusto ninyo og modern financial tools, Hybrid fits perfectly.

Mao nga architecture selection is a fit decision, dili kini popularity contest. On-premise concentrates control locally; cloud delegates the platform to a provider; hybrid handles both pero nagpataas sa integration complexity. Before choosing, management must always ask: What risks can we manage, and who owns each responsibility?

Transition

"A deployment choice is only the beginning. Karon, atong sutaon ang connected management considerations nga mag-determine kung magpabilin ba nga dependable ang inyong architecture sa actual operations.

[Clicks next slide]

Slide 13 - Architecture Challenges and Management Considerations

Stage Directions

[Shows Slide 13]

[Reveals the six cards]

[Draws attention to the connections among the cards]

[Instructor reads slide]

Verbatim Teacher Script

Architecture quality depends on technical and managerial decisions. These six considerations—security, availability, performance, scalability, integration, and governance—dili ni sila isolated problems, ha? They form one single system of responsibilities.

Let's look at how they connect:

- **Security:** This covers identity, authorization, and data encryption. Remember: the presentation tier authenticates, the application tier authorizes, and the data tier restricts access. Maski usa lang ka weak connection, kini makadaot ug maka-undermine sa tibuok secure server.
- **Availability:** Essential ERP services must remain usable. Gi-apil ani ang redundancy, backup, ug recovery. Ang organization gihapon ang mag-determine which processes are critical, ug kung unsa kadugay sila maka-afford nga ma-down ang system.
- **Performance:** Refers to response time and throughput. Ang network latency, database design, ug heavy transaction volume—kani sila ang makapahinay sa dagan. So eh trace sa jud ninyu ang full path first before ninyo eh blame ang user interface.
- **Scalability:** The ability to handle more users, data, and branches. Ang growth, dili lang na basta-basta kay mag-add lang og hardware—your workflows and configurations must scale too.
- **Integration:** Requires clear ownership, data definitions, and error handling. Ang interface dili pa mana human human human pagkahuman sa inyong first successful message; it must be monitored constantly habang nagasige og kausab ang mga systems.
- **Governance:** Kani ang nagakonektar sa tanan. It defines decision rights, vendor oversight, configuration standards, ug change control para magpabilin nga luwas ug well-documented ang mga updates. Sa practical side, let's see kung unsay mahitabo if ang inyong organization mag-add og bag-ong branch:

Dili lang kini basta-basta 'add one more user,' ha? Scalability requires new network capacity and roles. Sa Security naman, kailangan masiguro nga ang bag-ong team naay saktong access limits. Ang Performance, dapat magpabilin gihapon nga stable maski naay extra workload. Integration needs new branch codes, ug ang Governance pod, kinahanglan mag-approve ug mag-log sa tibuok configuration change. Tan-awa, usa lang ka decision pero ma-touch ang tibuok unom ka areas!

The big management lesson here is that technical quality depends on clear ownership. Every component, interface, and recovery process needs a responsible role. Kay kung lubog ang ownership, magpatong-patong gyud ang inyong risks.

Transition

"We have examined layers, deployment, and management challenges. Sa sunod nga slide, atong i-combine kining tanan uban ang mga tawo, mga proseso, controls, ug organizational goals.

[Clicks next slide]

Slide 14 - Architecture as an Integrated Organizational System

Stage Directions

[Shows Slide 14]

[Reveals People and Processes]

[Points to the ERP Architecture core]

[Reveals Controls and Organizational goals]

[Highlights the four information chips]

[Instructor reads slide]

Verbatim Teacher Script

ERP architecture aligns organizational design with technical structure. Kani nga slide highlights ngano nga importante gyud ang system architecture. Technology is just the foundation, pero it must always serve real organizational work.

As you can see, matching gyud ang technical core sa upat ka critical business areas:

- **People:** Sila ang nagdala sa roles, skills, ug accountability. The system needs to make responsibilities clear—ang users kinahanglan naay saktong screens, ug ang data owners naay right authority to maintain master records.
- **Processes:** Kini ang flow sa inyong trabaho. Ang application tier can only support a process nga klaro na daan ug na-configure og tarong. Always remember: if you automate a confused process, gipakuripot ra ninyo ang gubot.
- **Controls:** Basehan kini sa security, validation, ug auditability. Kinahanglan built-in na kini daan sa inyong architecture, dili kay kung kanus-a nahuman ang system, ayha pa huna-hunaon.
- **Organizational Goals:** Ang inyong business priorities ang naghatag og direction. Bisag unsa pa ka nindot or ka-high tech inyong structural design, bad fit gihapon na kung dili mosuporta sa inyong corporate goals.

Kani pong upat ka chips sa ubos summarizes our operating principles: **Shared data** keeps references consistent. **Integrated technology** allows controlled communication. **Managed responsibility** avoids finger-pointing kay naay tag-iya ang matag component. Ug **Continuous alignment** keeps the system updated habang nagka-change ang kumpanya.

Dili ni isolated responsibility, ha? If our people do not understand their roles, looking at a beautiful screen will not create accountability. Architecture fit is only achieved kung ang inyong technical design responsive ug nagasuporta sa pamaagi kung unsaon ninyo pag-operate ang inyong negosyo.

Transition

We'll end our session by integrate our three-tier structure, deployment options, ug ang matching logic sa organizational fit.

[Clicks next slide]

Slide 15 - Lesson Summary

Stage Directions

[Shows Slide 15]

[Reveals the three summary cards]

[Highlights the final callout]

[Instructor reads slide]

Verbatim Teacher Script

To wrap up our session, let's integrate the three key concepts we covered today:

1. **The Three-Tier Structure:** The **presentation tier** handles user interaction, the **application tier** processes the business logic, and the **data tier** maintains our shared records. Downward flow is the request; upward flow is the response.
2. **Deployment Choices:** On-premise focuses on direct local control and maintenance. Cloud gives you speed and delegates the platform to a provider. Hybrid blends both, requiring tight integration and synchronization rules. Remember, no model is universally superior; **fit** depends on your organization's resources, connectivity, and goals.
3. **Architecture Fit:** A sound design connects technology with people, processes, controls, and business objectives.

Ang main takeaway mao ni: ERP architecture gives you the technical foundation for integrated operations, but the actual value depends on **disciplined management and clear ownership** across the entire system.

Transition

Session 3 explained the technical structure behind ERP integration. In Session 4, we will move from the technical view to the functional view and examine the Finance, Procurement, and Inventory modules operating within that structure.

[Clicks next slide]

Session 3 Closing

Thank you, class. Before ta mag-break, naa bay question about the three tiers, the request flow, or the difference among on-premise, cloud, and hybrid deployment?

[Waits briefly for questions]

If there are no questions for now, remember this simple sequence: users interact, applications process, and the database preserves the controlled record. Ang organization then decides where those responsibilities will run and who will manage them.

Sa practical side, mao ni ang angay bantayan. For our next session, we will use this architecture as the background. We will not redefine all the layers again. Instead, we will examine what major ERP modules actually do inside the platform. Specifically, we will connect Finance, Procurement, and Inventory through shared data and one business process.

Pwede ninyo take a short break. Please return with the presentation still accessible, and we will continue with Session 4. Thank you.